LYU JIAXIN

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SUMMARY

MSc Statistical Data Science (UCD) seeking biostatistics roles. Re-analysed a retracted imaging study for my thesis, designed a survival-analysis simulation study in R, and built tools that turn published evidence into usable output. Almost two years of experience in product delivery.

EDUCATION

University College Dublin (UCD), MSc Statistical Data Science *Sep 2025 – Present*
Courses: Survival Models, Mathematical Statistics, MCMC, Machine Learning, Advanced Predictive Analytics, Bioinformatics, Data Programming with R

Harbin University of Commerce, BSc Statistics *Sep 2019 – Jun 2023*

PROJECTS

- **KETO-CTA Re-Analysis (MSc thesis; R).** Re-analysed the released data of a retracted imaging study (100 participants, 2 CT scans): compared regression models by AIC, checked them with log-scale and change-score sensitivity analyses and diagnostics (HC3, robust regression), and rebuilt a missing variable from a published plot. The data supported a qualified version of the original claim.
- **Survival Analysis Simulation (R, Quarto).** Designed with ADEMP; compared the Cox, log-rank and RMST tests under four effect patterns (24 scenarios × 1,000 runs). Null rejection 0.043–0.066 against a 0.05 target; with crossing curves, Cox power rose from 0.33 to 0.97 while RMST stayed near 0.05.
- **Check-up Planner (JavaScript).** Turned published screening guidelines (USPSTF and Chinese guidelines, 41 rules) into a bilingual tool that matches a person to check-up items, each labelled with its evidence level.
- **Scatter Plot Digitiser (Python, OpenCV, Streamlit).** Recovered missing thesis data from a published plot and validated it against known values (largest error 1.07 mm^3 on an axis range of about 160 mm^3).
- **Human Activity Recognition (R, Keras).** Compared a DNN and a CNN on wearable-sensor time series (19 activity classes); the CNN generalised better and reached 97.8% test accuracy.
- **EHS Management Product.** Benchmarked five vendors (about 196 core functions) and compared three systems to design a standard safety-management product, then tailored it to one customer's processes.

EXPERIENCE

Chengdu VisionVS Technology Co., Ltd., Product Assistant & Delivery Engineer, Chengdu, China *Jul 2023 – Mar 2025*
Products: interactive electronic technical manuals (IETM, IETP) and a safety production management (EHS) system (SPM).

- Researched the market and competitor products; forecast market size and revenue to support business-model evaluation.
- Led an internal project-management system from a survey of four existing tools through requirements, process design and delivery.
- Defined functional requirements with clients and wrote design specifications, project plans and technical documentation.
- Delivered product training and worked with developers on iterations based on user feedback.

SKILLS

Programming & stats: R, Python, SQL, SAS

Methods: regression modelling, survival analysis, simulation study design, sensitivity analysis, machine learning, deep learning (CNN, DNN; Keras in R)

Tools: Excel, Figma, Axure, Git/GitHub, Quarto, Markdown, UML

Bioinformatics (UCD coursework): NCBI protein BLAST, Clustal Omega alignment and phylogenetic trees (SARS-CoV-2 spike homologues, grouped with SARS-related and bat coronaviruses); alignment and tree-building algorithms (Needleman-Wunsch implemented in R; Smith-Waterman, BLAST, WPGMA, neighbour joining); protein structure prediction; GenBank, UniProt, PDB

Languages: English (IELTS 6.5 overall), Mandarin (native)